Enhancing Edge Computing with Federated Learning and AI

As AI moves to the edge, decentralised model training and federated learning will reshape privacy and efficiency paradigms. The integration of peripheral AI with 5G technology will enhance the capabilities of devices on the edge.

The convergence of edge computing and artificial intelligence (AI) has resulted in the emergence of a paradigm shift referred to as edge AI. This novel methodology seeks to deliver the computational capabilities of AI directly to edge devices, including but not limited to smartphones, IoT devices, and other interconnected electronics. The integration of AI and edge computing facilitates instantaneous data processing, analysis, and decision-making at the network's periphery, presenting a multitude of benefits in comparison to conventional cloud-centric AI frameworks.

Edge AI pertains to the implementation of models and algorithms for artificial intelligence that are executed directly on edge devices or local edge servers. By obviating the necessity to transmit data to a centralised cloud server for processing, edge AI effectively diminishes latency and optimises efficiency. By capitalising on the computational capabilities inherent in the devices, this methodology expedites response times and enhances the overall user experience.

Edge computing in AI

With the increasing need for real-time decision-making and low-latency applications, cutting-edge AI solutions are significantly contributing to the